**B. TECH-ME-III-I Sem**

| 23A03504a | TOOL DESIGN<br>(Professional Elective-IV) | L | T | P | C |
|---|---|---|---|---|---|
| | | 3 | 0 | 0 | 3 |

| **Course objectives:** The objectives of the course are to | |
|---|---|
| 1 | **Understand** the fundamentals of tool engineering and the role of tool design in manufacturing. |
| 2 | **Analyze** the principles of metal cutting and apply them to cutting tool design. |
| 3 | **Design** various jigs and fixtures using proper locating and clamping principles. |
| 4 | **Evaluate** and design different types of press tool dies for sheet metal operations. |
| 5 | **Develop** tooling and fixture strategies suitable for CNC machining systems. |

| **Course Outcomes:** On successful completion of the course, the student will be able to, | | |
|---|---|---|
| 1 | Understand tool design fundamentals, select appropriate materials, and design effective tools to develop durable, precise tools for various manufacturing applications. | L2, L3,L6 |
| 2 | **Define** Oblique and orthogonal cutting **, Apply** the mechanics of metal cutting to design basic cutting tools like single-point, milling, and broaching tools. | L1,L3,L6 |
| 3 | **Demonstrate basic principles of** drill jigs and various fixtures and design the jigs and fixtures by applying principles of location and clamping. | L2,L3,L6 |
| 4 | **Calculate** clearance, cutting forces, and **develop** designs for press tool dies (blanking, piercing, bending, and drawing). | L2,L4,L6 |
| 5 | **Evaluate and Develop** tool holding, fixture systems, and automation features like ATC for CNC machine tools. | L1,L5,L6 |

**UNIT I**

**INTRODUCTION TO TOOL DESIGN**

Introduction –Tool Engineering – Tool Classifications– Tool Design Objectives – Tool Design in manufacturing- Challenges and requirements- Standards in tool design-Tool drawings -Surface finish – Fits and Tolerances - Tooling Materials- Ferrous and Non ferrous Tooling Materials- Carbides, Ceramics and Diamond -Non metallic tool materials-Designing with relation to heat treatment.

**UNIT II**

**DESIGN OF CUTTING TOOLS**

Mechanics of Metal cutting –Oblique and orthogonal cutting- Chip formation and shear angle - Single-point cutting tools – Milling cutters – Hole making cutting tools- Broaching Tools - Design of Form relieved and profile relieved cutters-Design of gear and thread milling cutters.

**UNIT III**

**DESIGN OF JIGS AND FIXTURES**

Introduction – Fixed Gages – Gage Tolerances –selection of material for Gauges – Indicating Gages – Automatic gages – Principles of location – Locating methods and devices – Principles of clamping – Drill jigs – General considerations in the design of drill jigs – Drill bushings – Methods of construction –Types of Fixtures – Vice Fixtures – Milling Fixtures – Boring Fixtures – Broaching Fixtures.

**UNIT IV**

**DESIGN OF PRESS TOOL DIES**

Types of Dies –Method of Die operation–Clearance and cutting force calculations- Blanking and Piercing die design – Pilots – Strippers and pressure pads- Presswork materials – Centre of pressure -Strip layout – Short-run tooling for Piercing – Bending dies – Drawing dies-Design and drafting.

**UNIT V**

**TOOL DESIGN FOR CNC MACHINE TOOLS**

Introduction –Tooling requirements for Numerical control systems – Fixture design for CNC machine tools-Sub plate and tombstone fixtures-Universal fixtures– Cutting tools– Tool holding methods– Automatic tool changers and tool positioners – Tool presetting– General explanation of the Brown and Sharp machine.

**Textbooks:**

1. Cyrll Donaldson, George H.LeCain, V.C. Goold, “Tool Design”, Tata McGraw Hill Publishing Company Ltd., 2000.
2. E.G.Hoffman,” Jig and Fixture Design”, Thomson Asia Pvt Ltd, Singapore, 2004.

**Reference Books:**

1. P.C.Sharma, A Text book of Production Engineering, S.Chand Publications, 1999.
2. Prakash Hiralal Joshi, “Tooling data”, Wheeler Publishing, 2000
3. Venkataraman K., “Design of Jigs, Fixtures and Presstools”, TMH, 2005.
4. Haslehurst M., “Manufacturing Technology”, The ELBS, 1978.

**Online Learning Resources:**

- https://www.iare.ac.in/sites/default/files/lecture_notes/TOOL%20DESIGN_Lecture_Notes.pdf
- https://www.cet.edu.in/noticefiles/261_MMP%20Lecture%20Notes-ilovepdf-compressed.pdf
- https://www.vssut.ac.in/lecture-notes.php?url=production-engineering
- https://nptel.ac.in/courses/112/105/112105233/
- https://www.youtube.com/watch?v=7MkX-sW97rI
- https://nptel.ac.in/courses/112/105/112105126/#